https://github.com/Et1enneee/comp3500-logistics

COMP 3500SEF · Final Report

Warehouse Logistics Management System
Group 42: Tang Yi Jie, Yin Pei Xu, Tang Chao, Yang Ting Zhang, Wu Peng Bo, Chung Ngon Man, Li Mu Qiu.

1. Scope

The goals are traceable order decisions and a role-separated workflow. Customers order and track; staff confirm, pick, ship and assign; drivers simulate assigned deliveries; administrators report. The local application serves one warehouse using Next.js, TypeScript, Prisma and SQLite. Real GPS, barcode scanning, recipient telephone collection, payment and carrier integration are outside scope.

2. Development process

Plan-driven incremental development defines each increment by its trigger, scope and exit criterion; integration and continuous integration provide feedback. The records establish development, not external delivery.

Table A. Increment, trigger and outcome

Increment and date	Trigger	Outcome
Planning, 9 and 11 Oct 2026	Project selection	Responsibilities and contracts recorded
Test-led review, 17 Oct 2026	Assess order and stock functions	Defects recorded
Confirmation, 26–28 Oct 2026	Failed endpoint and missing ledger	Transaction verified
Integrated application, 1 Nov 2026	Integrate foundation	Customer, staff and admin workflows
Driver and privacy, 2 Nov 2026	Delivery and data minimisation	Assigned tasks and privacy checks
Documentation hardening, 10–11 Nov 2026	Model and evidence review	Diagrams and traceability aligned

Git manages versions, the workflow integrates changes, and the issue tracker records problems. Increment evidence.

3. Requirements

The software requirements specification is the specification. Functional requirements cover FR-01 order creation; FR-02 state changes; FR-03 numbering and tracking; FR-04 inventory and alerts; FR-05 notifications; FR-06 reports; FR-07 access control; FR-08 simulated delivery. Non-functional requirements cover NFR-01 performance; NFR-02 integration-point availability; NFR-03 least privilege; NFR-04 scripted operation effort; NFR-05 process quality.

Table B. Risk coverage (basis and verification)

Risk	Addressed by	Register status
RK-01	Process	Open
RK-02	FR-02, NFR-05	Mitigated
RK-03	Process	Open
RK-04	NFR-05	Open
RK-05	FR-01/04/06, NFR-05	Open
RK-06	Process	Open
RK-07	NFR-05 support	Open
RK-08	Process	Open

4. System models

Layered architecture separates pages, routes and authentication, business handlers, and persistence. State and stock guards and transaction atomicity enforce hard rules.

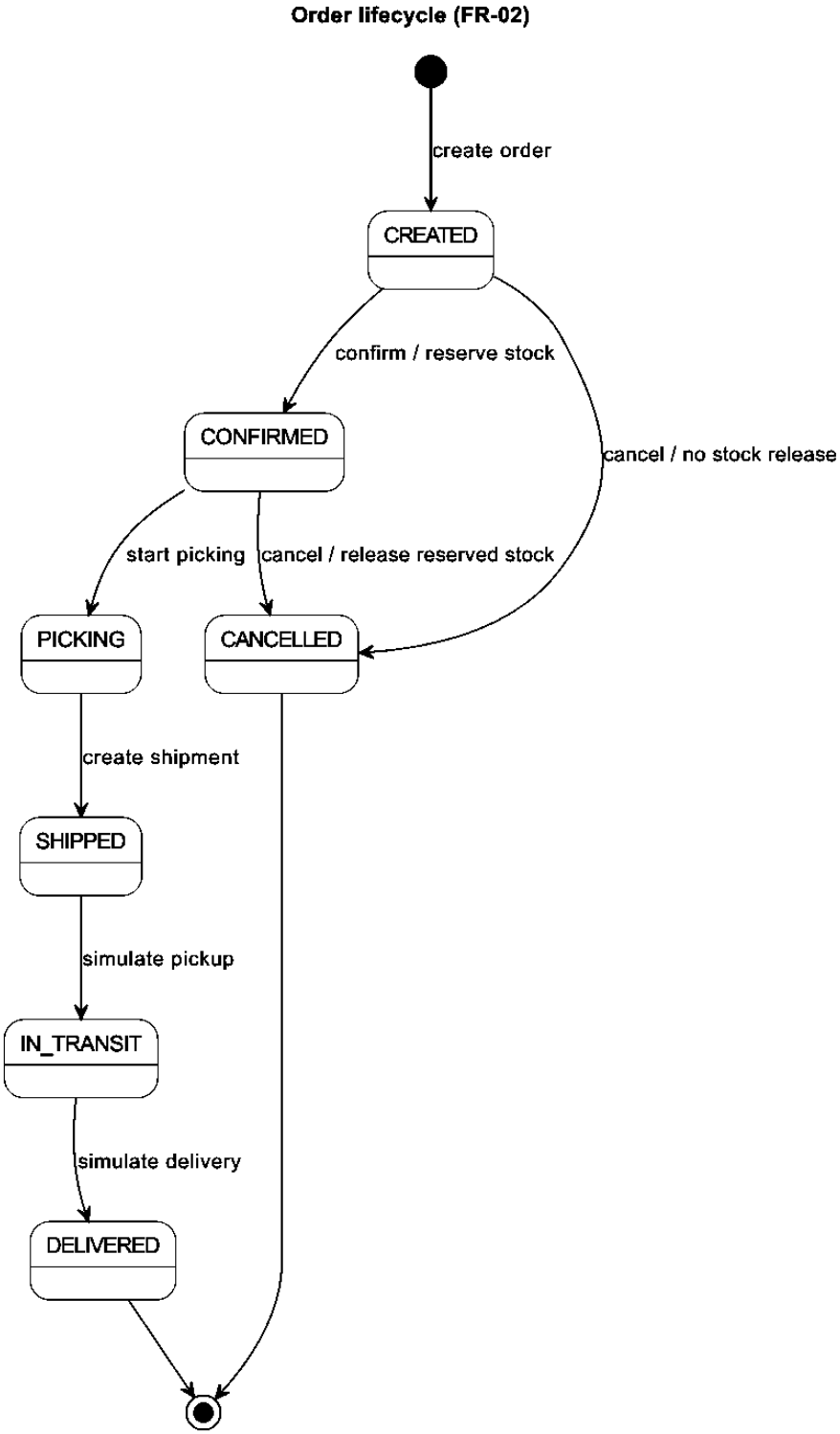

Figure 1. Order lifecycle: valid transitions and effects. Source.

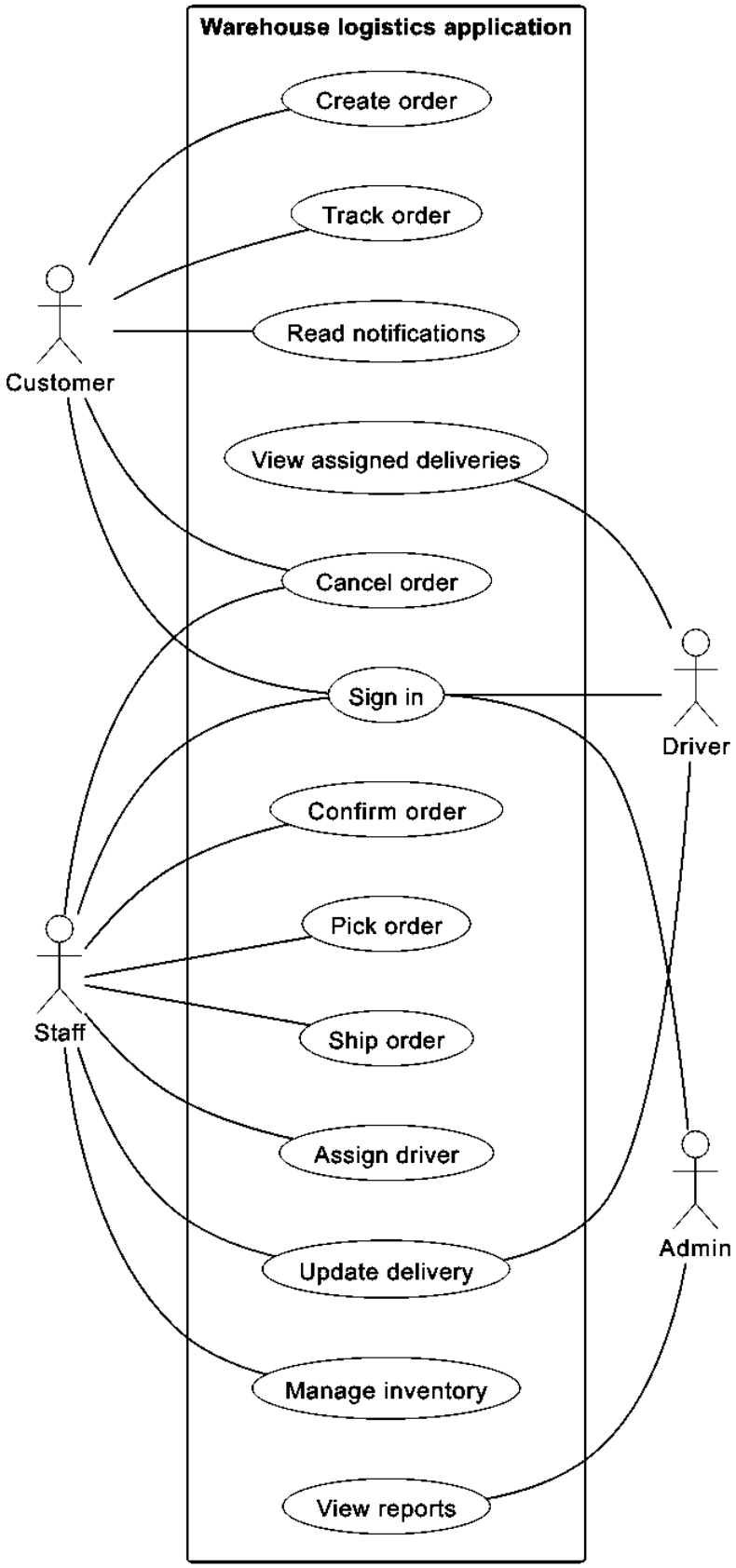

Figure 2. Use cases associate external roles with application functions. Source.

An order has 0..* items in the schema, while creation requires at least one; an item references one product. An order has 0..1 shipment; a product has 0..* inventory logs.

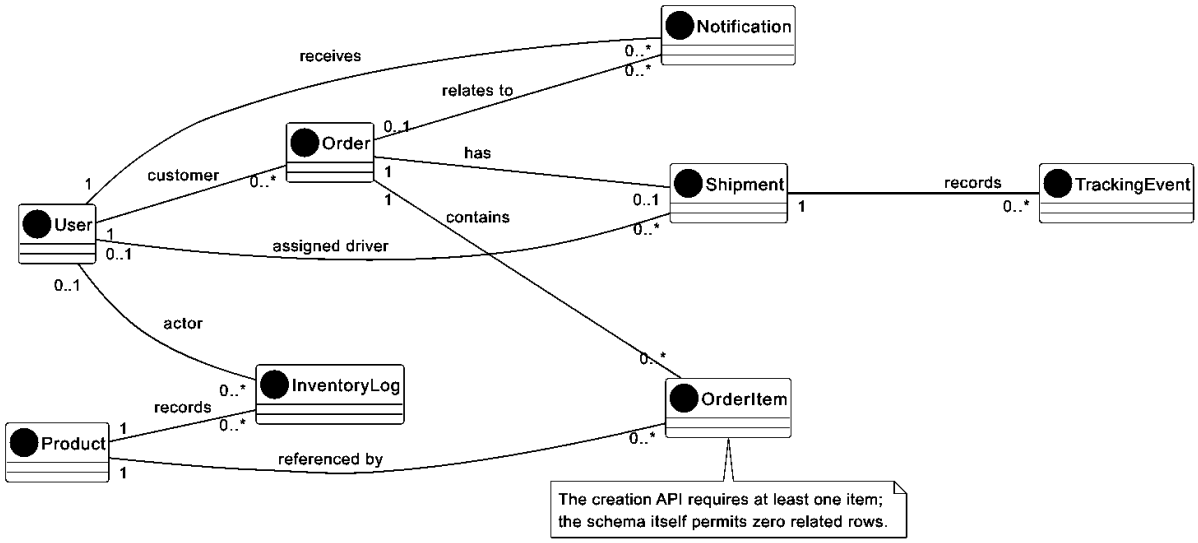

Figure 3. Entity relationships and multiplicities. Source.

5. Implementation

The confirmation guard (order-api.ts:250–253) requires CREATED; the stock guard (order-api.ts:276–280) requires sufficient quantity. The serializable transaction groups state, stock, ledger and notification writes. Staff cancellation requires CONFIRMED.

6. Verification and validation

Verification checks the specification; validation assesses fitness for real users. The recorded run passed 38 unit, 140 integration, 2 delivery and 25 browser checks; logs and CI provide evidence. Repeated confirmation and insufficient stock return HTTP 409 with unchanged state and stock. On local loopback, API p95 was 3.2–13.6 ms and first-contentful-paint p95 was 64–96 ms (measurement). The revised availability criterion records 40 successful build–test–browser integration points; continuous sampling supplies supplementary evidence (observations).

7. Evolution

The early defect record led to the confirmation transaction and its retest. The post-presentation audit led to corrected models and renewed tests. The scripted walkthrough exposed a four-action administrator task, tracked in issue #15.

8. Validation

The automated walkthrough counts at most three scripted actions for each of ten selected operations from prepared screens. The human protocol specifies independent role tasks, completion, errors, assistance and time as supplementary validation; the evidence records no participant results.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback	Disposition	Evidence
Diagrams	Corrected state and use-case models	Audit
FR-04	Stock transition effects belong to FR-02	Traceability
Increments	Defined triggers and outcomes	Register
Presentation p95	Current probe governs report	Measurement
Admin task	Stronger whole-task target tracked	Issue #15

The matrices and increment records connect decisions to implementation and evidence.

10. Evidence

Traceability · Specification · Evidence directory · Baseline record.

Full evidence trail: docs/final-report.md in the repository (evidence baseline `d5ce34e8b57e9d318808d214f6ac36ceb7ea0198`).